



Assessment Cover Sheet

Assessment Title	Group Project		
Assessment Type	Uncontrolled	Group	Not must-pass
Due Date	02/07/2026	Course Code	IT8415
Course Title	Database Programming 2		
Internal Moderator's Name			
External Examiner's Name			

Instructions:

1. This cover sheet must be completed (section in blue below) and attached to your assessment before submission in hard copy/soft copy.
2. This assessment carries 100 marks assessing CILO 1, 2 and CILO 3.
3. The **use of generative AI tools is strictly prohibited**.
4. References consulted (if any) must be properly acknowledged and cited.
5. The assessment has a total of 12 pages.

Learner ID	XXXXX	Date Submitted	Xx/xx/xxxx
Learner Name	XXXXX		
Programme Code	IT8415		
Programme Title	Database Programming 2		
Lecturer's Name			

By submitting this assessment for marking, I affirm that this assessment is my own work.

Do not write beyond this line. For assessor use only.

Learner Signature

Assessor's Name

Marking Date

Marks Obtained

Comments:

Contents

Database Programming 2 Group Project: General Information:	3
Learning Outcomes Assessed:	3
AI Policy:.....	3
Project Instructions:	4
Use of Software	4
Deliverables.....	4
Assessment Description:	5
1- Project Title: Data-Driven Web Application (Student-Selected Theme)	5
Task 1- Functional Requirements.....	6
1.1 User Roles & Authentication.....	6
1.2 Main Page (Home Page).....	6
1.3 Search Feature	7
1.4 Comments & Rating System	7
Task 2- Non-Functional Requirements	8
Task 3 – Database Requirements.....	8
Task 4 – Advanced Features	8
Task 5 – Test Plan.....	8
2- Individual Report.....	10
3- Group Project Presentation.....	Error! Bookmark not defined.
Marking Scheme:	11

Database Programming 2 Group Project: General

Information:

Project Submission Date: Tuesday, 02 June 2026 by 11:59 p.m.

Assessment Type: Group Project – Maximum 5 members per group.

Learning Outcomes Assessed:

1. Critically apply specialist-level knowledge and techniques to design, develop, and test data-driven, server-side applications.
2. Apply professional standards and conventions with creativity to ensure the effective development of software solutions.

AI Policy:

Use of AI tools is permitted to review and refine outcomes from standard practices taught in the degree and practiced in the profession. Students can use it for general purposes of improving academic skills, developing digital and AI literacy skills, and enhancing the overall teaching-learning process. Examples of the above include, but are not limited to:

- 1) Brainstorming and initial ideation for critical examination and evaluation.
- 2) Production of summaries or syntheses.
- 3) Comprehension checks for the meanings of complex terms or concepts.
- 4) Analysis of data, contents, and recurring patterns.
- 5) Creation and enhancement of images and other artworks.
- 6) Creation of codes.

Students should consult with their supervisor when using AI tools to receive feedback on best practices.

AI may be used for literature summaries and grammar polishing (as directed by the tutor), but students must fully understand and explain the outputs. AI should not be used for thesis writing, analysis, or literature reviews, and all references must be authentic. Any AI assistance must be acknowledged in the methodology or acknowledgments section and should be properly cited.

Project Instructions:

Instructions, use of software and deliverables are given in the sub-sections below.

Use of Software

The project must be presented in a professional manner.

All the tasks should be produced by using all the software taught within the course, PHP and MySQL for database. The student can choose their preferred coding environment.

Deliverables

Group Leader Submission:

All files must be placed in a single ZIP file named:

IT8415_GroupName_LeaderID.zip

The Word document must be converted to PDF.

Contents of ZIP:

- PDF document containing:
 - Application URL (Server deployment only)
 - Testing accounts (Admin, Creator, Viewer)
 - ERD screenshot
 - GitHub repository link
 - Advanced features explanation
 - Complete Test Plan
- NetBeans project files and folders
- SQL scripts (tables, procedures, triggers if applicable)

Individual Submission:

- Individual reflection on the work done and the overall group communication and cooperation. The word limit is 200 - 250 words (-/+ 10%).

Assessment Description:

1- Project Title: Data-Driven Web Application (Student-Selected Theme)

You are expected to develop a web application using Apache, PHP and MySQL. Apply the knowledge and practice studied in the class. The application must be data-driven and demonstrate server-side programming, database design, validation, testing, and professional development practices.

Each group may select their own application topic, subject to tutor approval.

Examples of Possible Application Themes:

- Blog or Content Publishing Platform
- Online Learning Portal
- Movie Review System
- E-Commerce Mini Store
- Event Management System
- Job Portal
- Travel & Tourism Guide
- Book Review Platform
- Fitness Tracking System
- Recipe Sharing Platform
- Or any other approved idea

Regardless of the selected topic, all projects must implement the required functional, non-functional, database, and testing requirements described below.

Task 1- Functional Requirements

1.1 User Roles & Authentication

The system must support at least three types of users:

- Visitor / Viewer
- Content Creator / Contributor
- Administrator

Login is required for content creation and management. Visitors may browse/search content without logging in (unless your project logic requires otherwise).

Mandatory:

- Sign-up and Login functionality
- Session management
- JavaScript-based validation
- Server-side validation
- Encrypted passwords

1.2 Main Page (Home Page)

The system must include:

- Navigation menu for categories/sections
- Search functionality
- Reverse chronological listing of content (newest first)

Each item must include:

- Title
- Short description
- “View More” link
- At least one image
- At least one media file (audio/video optional depending on theme)
 - Optional downloadable content

1.3 Search Feature

Must include search by:

- Title (engine-based search)
- Date or date range
- Creator/Author/etc
- Popularity (rating/view count)

Full-text or indexed search is required.

1.4 Comments & Rating System

- Users can rate content (like/dislike or star rating)
- Logged-in users can add comments
- Admin can remove inappropriate comments
- Viewers able to see comments

1.5 Creator Panel

Content creators must be able to:

- Add new content
- Edit content before publishing
- Upload media
- Publish content
- View their own content

Once published, editing rules can be restricted based on your design logic.

1.6 Admin Panel

Administrator must be able to:

- Manage users
- Manage all content
- Remove content (with system message shown)
- Generate reports (e.g., most popular content, content by creator)

1.7 Reporting System

Admin must generate at least two reports:

- Most popular content within a date range
- Content created by a specific user

Task 2- Non-Functional Requirements

1. Professional UI design
2. Responsive layout
3. Consistent navigation
4. Pagination if content > 10 records
5. Proper validation
6. Secure DB usage
7. Test data sufficient for demonstration

Task 3 – Database Requirements

1. ERD required
2. At least one stored procedure
3. At least 3 user roles
4. Minimum test dataset (you may generalize minimum 15 records total, with at least 5 in a unique category)
5. All tables must follow a fixed prefix rule (e.g., dbProj_)

Task 4 – Advanced Features

Use at least 2:

- AJAX
- Localization
- Triggers
- Prepared statements
- JQuery
- Advanced UI
- Others (approved)

Task 5 – Test Plan

Each group must create a comprehensive Test Plan document.

This document must include:

- Screenshots of all pages of the web application

- Step-by-step instructions explaining how to test every functional requirement
- Step-by-step instructions explaining how to test each advanced feature
- Clear demonstration that each requirement works correctly

Word Limit Requirements:

- Each functional requirement explanation must not exceed **350 words**
- Each advanced feature explanation must not exceed **350 words**
- No limit on the number of screenshots

What the Test Plan Must Demonstrate:

For every functionality:

1. The purpose of the feature
2. Steps to test it
3. Expected output
4. Screenshot evidence
5. Any validation or error handling shown

The Test Plan should prove that the system:

- Works correctly
- Handles invalid input properly
- Enforces role permissions
- Produces correct reports
- Implements advanced features successfully

The Test Plan may be submitted in PDF format.

2- Individual Report

Each student must submit an individual reflection.

Requirements:

- Word limit: 200 - 250 words ($\pm 10\%$)
- Could include:
 - Personal contribution to the project
 - Challenges faced
 - Problem-solving approach
 - Group communication and cooperation
 - Skills learned (technical and teamwork)
 - Areas for improvement

This reflection should demonstrate professional awareness and self-evaluation of learning outcomes.

Marking Scheme:

Project Components	Marks
Generic: • Cover page • GitHub link • Overall report quality and structure • Meeting all requirements	3
Application functionality • Functional Requirements • Nonfunctional Requirements	55
Database • ERD • Procedure • Index Searching & Full text searching • DB security technique used. • SQL Scripts	20
Advanced Features	12
Test Plan	10